

Hyeonjae Park

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Education

Georgia Institute of Technology | Atlanta, GA

Master of Science in Electrical and Computer Engineering, GPA 3.62/4.00
Bachelor of Science in Computer Engineering, GPA 3.62/4.00 (Institute Highest Honors)

January 2025 – Present
Expected Graduation, May 2026
December 2024

Skills

Programming: Python, C/C++, C#, MATLAB, Java, JavaScript, HTML, SQL

Frameworks & Tools: ROS/ROS2, PyTorch, TensorFlow, OpenCV, OpenGL, CUDA, Docker, Git, Linux, GoogleTest, RTOS

Simulation: MuJoCo, NVIDIA Isaac Sim, Gazebo

Robotics: Reinforcement/Imitation Learning, Behavior Cloning, Motion Planning, Nonlinear Control, State Estimation, Kinematics

Hardware: Jetson, ESP32, Raspberry Pi, Arduino, Pixhawk, ARM Mbed, oscilloscope, Onshape, Fusion 360, Solidworks, Eagle CAD

Languages: Korean (native), English (fluent), Chinese (beginner)

Experience

Georgia Institute of Technology | Atlanta, GA

March 2024 – Present

Graduate Research Assistant / Safe Robotics Lab

- Led development and deployment of coach-student reinforcement learning framework in MuJoCo, implementing SAC/PPO-based shared control system on a custom assistive unicycle platform with gamepad teleoperation in Python.
- Developed GRU-based skill estimator and behavior cloning pipeline in PyTorch on 730K-sample dataset, reducing prediction MSE by 49% (0.188 to 0.096) over 100 epochs to accelerate policy convergence.
- Executed 20+ human subject experiments validating deployed framework, preparing first-author manuscript for IEEE RA-L.

Georgia Tech Research Institute | Atlanta, GA

May – August 2025

Robotics Research Intern / ISTD - ATAS

- Developed a high-fidelity digital twin of a custom 14-DOF dual UR5e industrial manipulator cell in NVIDIA Isaac Sim, building ROS2 control stack with Docker for real-time sim-to-real deployment.
- Built compliance controller in C++ for industrial assembly applications (peg-in-slot, sliding), achieving 100% task success rate.
- Achieved $\leq 0.684^\circ$ joint and $\leq 0.001\text{m}$ carriage accuracy across 5 hardware runs; presented at GRIP 2025 Poster Symposium.

Georgia Institute of Technology | Atlanta, GA

October 2024 – May 2025

Graduate Researcher / Dynamic Mobility Lab

- Designed bio-inspired multi-link foot mechanism with Bezier curve geometry for bipedal robot, optimizing linkage parameters for energy efficiency and gait stability.
- Migrated BRUCE humanoid robot to MuJoCo in C++, integrating a live dynamics visualization tool for real-time analysis.
- Built GoogleTest validation framework for automated regression testing of hybrid dynamics and ground contact transitions.

Projects

Autonomous Campus Navigation Drone | ECE, Capstone Design

Fall 2024

Team Lead

Led six-member team to design and deploy an autonomous quadcopter for campus navigation for visitors, securing \$2k in funding.

- Fabricated custom drone frame and PCB with 2D LiDAR and camera; validated flight system in Gazebo simulation with ROS.
- Designed global path planning with Dijkstra's algorithm and local obstacle avoidance using Artificial Potential Fields, smoothed with Bezier spline interpolation for collision-free flight.
- Enabled vision-based autonomous landing using OpenCV with PnP pose estimation, achieving $\pm 10\text{cm}$ landing precision.

Laser Lidar Turret | ECE, Embedded System Design

Fall 2023

Co-Team Lead

Team project for developing low-cost, modular autonomous targeting turret.

- Developed C/C++ control firmware on Mbed LPC1768 using Mbed RTOS, enabling a robust Bluetooth-controllable device.
- Implemented multithreaded embedded control system to ensure real-time LiDAR sensor processing and servo actuation.
- Integrated Raspberry Pi as high-level controller for motion detection, achieving tracking at 50Hz update rate.

Safe Path Planning with Eikonal PDE | ECE, PDEs Image Proc & Vision

Spring 2025

Individual project for solving optimal path planning problems in obstacle-rich environments.

- Implemented custom Fast Marching solver in MATLAB for Eikonal PDE with 8-connected grid traversal and exponential cost field to generate paths maintaining safe obstacle clearance.
- Validated on 3 environments across 48 parameter configurations, benchmarking path cost, lengths, and computation time.